mers — Extracted Transmission Models

Overview

A total of 101 transmission models were extracted from 46 articles that were flagged for containing extractable transmission models. The predominant model architecture was Compartmental (57.4%), with Human to human (direct contact) (20.8%) representing the most commonly modelled transmission route.

Summary

Metric	Value
Models extracted	101
Articles considered	46
Deterministic models	36 (35.6%)
Stochastic models	47 (46.5%)
Models with available code	31 (30.7%)

Figure 1: Model Overview

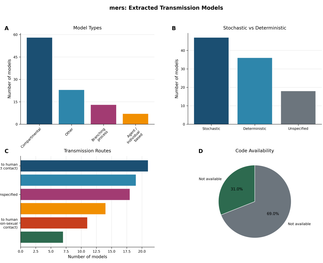

Figure 1. Overview of 101 transmission models extracted for mers. (A) Distribution of model architectures. (B) Classification by stochastic or deterministic formulation. (C) Primary transmission routes incorporated. (D) Proportion of models with publicly available source code.

Model Formulation

Deterministic formulations were employed in 35.6% of models, while stochastic approaches accounted for 46.5%. Empirical data were incorporated in 56.4% of the extracted models, with 37.6% classified as purely theoretical.

Figure 2: Interventions and Assumptions

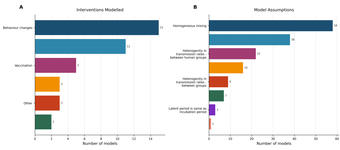

Figure 2. Interventions and assumptions in extracted mers transmission models (*n*=101). (A) Frequency of intervention types evaluated across models. (B) Common modelling assumptions. Categories are not mutually exclusive; individual models may incorporate multiple interventions or assumptions.

Interventions and Assumptions

The most frequently modelled interventions were Behaviour changes (14.9%), Quarantine (10.9%), Vaccination (5.0%). Common modelling assumptions included Homogeneous mixing (57.4%), Heterogeneity in transmission rates - between human groups (21.8%), Heterogeneity in transmission rates - over time (15.8%).

Figure 3: Programming Languages

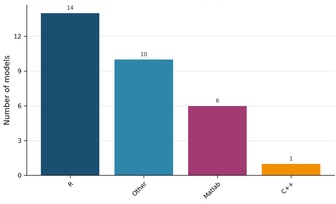

Figure 3. Programming languages used for model implementation among 31 models with specified language.

Figure 4: Model Type by Formulation

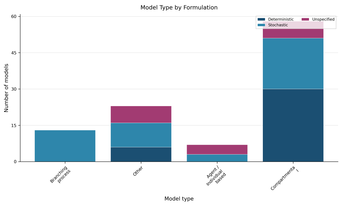

Figure 4. Distribution of deterministic and stochastic formulations across model types (*n*=101).

Reproducibility

Source code was publicly available for 30.7% of the extracted models.

Tables

Table 1: Model Types

Model Type	Count	Proportion
Compartmental	58	57.4%
Other	23	22.8%
Branching process	13	12.9%
Agent / Individual based	7	6.9%

Distribution of model architectures across 101 extracted transmission models.

Table 2: Model Formulation

Formulation	Count	Proportion
Stochastic	47	46.5%
Deterministic	36	35.6%
Unspecified	18	17.8%

Classification of extracted models by stochastic versus deterministic formulation (n=101).

Table 3: Transmission Routes

Transmission Route	Count	Proportion
Human to human (direct contact)	21	20.8%
Airborne or close contact	19	18.8%
Unspecified	18	17.8%
Human to human (direct contact);Vector/Animal to human	14	13.9%
Human to human (direct non-sexual contact)	11	10.9%
Vector/Animal to human	7	6.9%
Human to human (direct contact);Airborne or close contact	4	4.0%
Airborne or close contact;Human to human (direct contact)	3	3.0%
Human to human (direct non-sexual contact);Vector/Animal to human	2	2.0%
Airborne or close contact;Human to human (direct non-sexual contact)	1	1.0%
Airborne or close contact;Vector/Animal to human	1	1.0%

Primary transmission routes incorporated in extracted models (n=101).

Table 4: Interventions Modelled

Intervention Type	Count	Proportion
Behaviour changes	15	14.9%
Quarantine	11	10.9%
Vaccination	5	5.0%
Hospitals	3	3.0%
Other	3	3.0%
Contact tracing	2	2.0%

Types of interventions evaluated in extracted models. Categories are not mutually exclusive (n=101).

Table 5: Model Assumptions

Assumption	Count	Proportion
Homogeneous mixing	58	57.4%
Other	38	37.6%
Heterogenity in transmission rates - between human groups	22	21.8%
Heterogenity in transmission rates - over time	16	15.8%
Heterogenity in transmission rates - between groups	9	8.9%
Age dependent susceptibility	7	6.9%
Latent period is same as incubation period	3	3.0%
Heterogenity in transmission rates - between human and vector	1	1.0%

Common modelling assumptions. Categories are not mutually exclusive (n=101).

Table 6: Code Availability

Code Available	Count	Proportion
Yes	31	30.7%
No	69	68.3%

Availability of publicly accessible source code for extracted models (n=101).

Table 7: Programming Languages

Programming Language	Count	Proportion
Unspecified	70	69.3%
R	14	13.9%
Other	10	9.9%
Matlab	6	5.9%
C++	1	1.0%

Programming languages used for model implementation (n=101).

Table 8: Empirical Data Usage

Empirical Data Used	Count	Proportion
Yes	57	56.4%
No/Unspecified	44	43.6%

Model Inventory (Sample)

Table 9 presents a sample of 10 extracted model records. The complete dataset of 101 extracted transmission models for mers is publicly available.

Table 9: Sample of Extracted Models

Article ID	Model Type	Compartmental Structure	Formulation	Transmission Route
PMID_23787162	Branching process	Not compartmental	Stochastic	Human to human (direct contact);Vector/Anim
PMID_26713252	Other	Not compartmental	Deterministic	Unspecified
PMID_28960494	Agent / Individual based	Not compartmental	Unspecified	Airborne or close contact
PMID_29336306	Other	Not compartmental	Stochastic	Vector/Animal to human
PMID_29336306	Other	Not compartmental	Stochastic	Vector/Animal to human
PMID_29336306	Other	Not compartmental	Stochastic	Vector/Animal to human
PMID_29336306	Other	Not compartmental	Stochastic	Vector/Animal to human
PMID_29336306	Other	Not compartmental	Stochastic	Vector/Animal to human
PMID_29336306	Other	Not compartmental	Stochastic	Vector/Animal to human
PMID_29336306	Branching process	Not compartmental	Stochastic	Human to human (direct contact)

Sample of 10 records from 101 extracted mers transmission models.

Data Availability

The complete dataset of extracted transmission models for mers (*n*=101) has been made publicly available to support future modelling efforts and outbreak preparedness.